

Project Demonstration

Smart Lender: AI-Powered Loan Approval Prediction System

Demonstration Overview

The Smart Lender web application demonstrates how machine learning can be used to predict loan approval based on applicant details. The system provides a simple and user-friendly interface where users enter their information and instantly receive a prediction.

Terminal Execution

The Smart Lender application was executed successfully using the Flask development server. After running the application with the command below, the server started successfully and hosted the web application on the local machine.

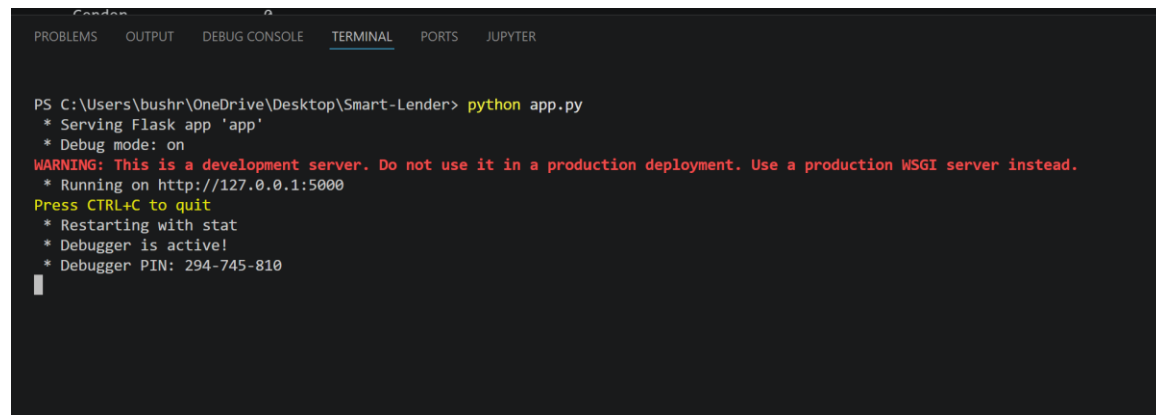
```
python app.py
```

The terminal displayed the following message, indicating that the application was running successfully:

```
* Running on http://127.0.0.1:5000
```

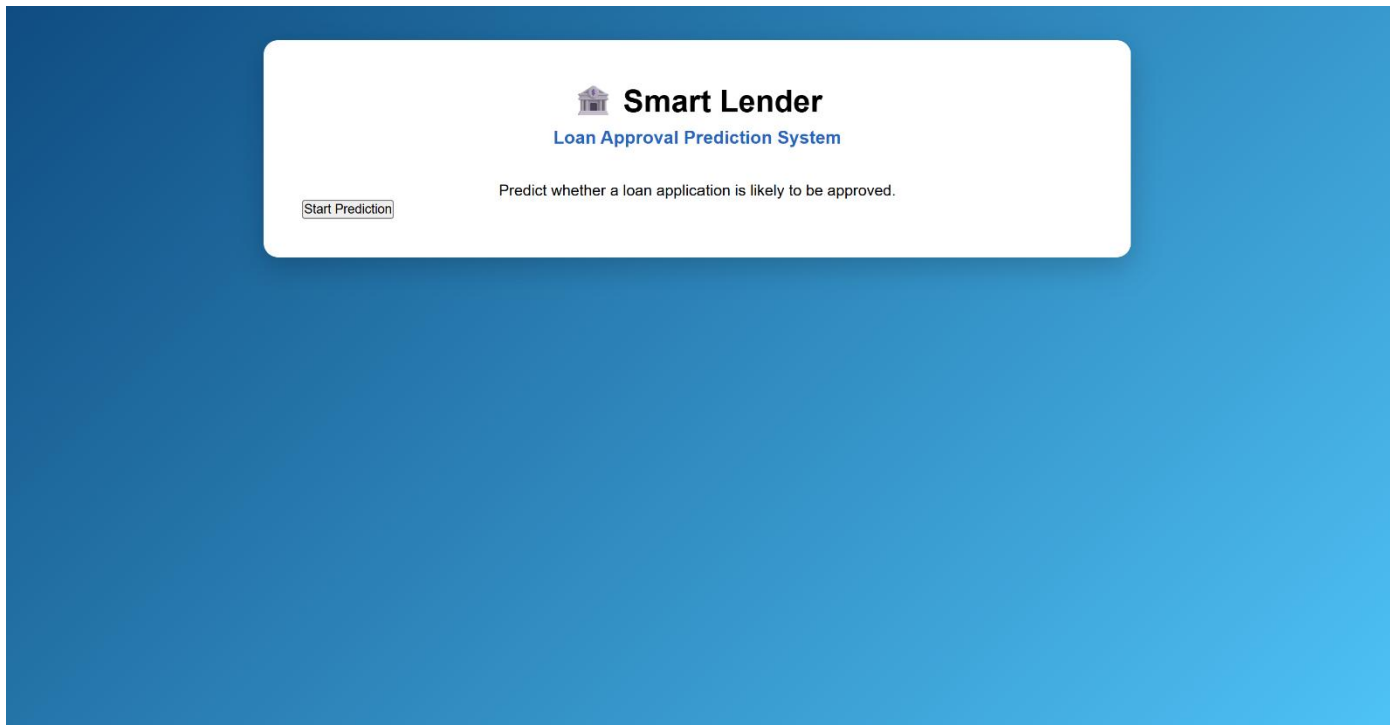
The local URL was then opened in a web browser to access the Smart Lender web application.

Flask Server Running in Terminal



```
PS C:\Users\bushr\OneDrive\Desktop\Smart-Lender> python app.py
* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
* Restarting with stat
* Debugger is active!
* Debugger PIN: 294-745-810
```

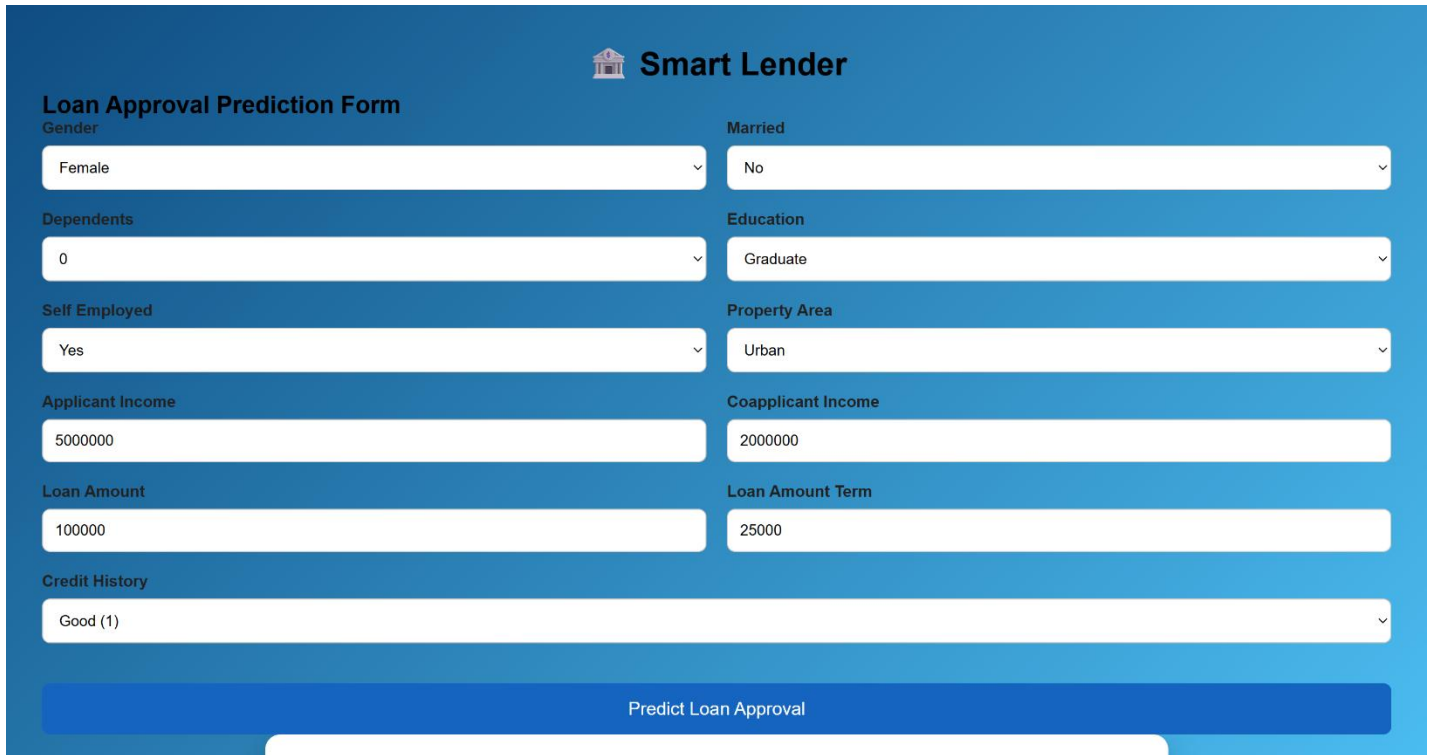
Picture 1 – Home Page



Description:

The home page welcomes users to the Smart Lender application. It provides a brief introduction to the Loan Approval Prediction System and includes a Start Prediction button that navigates users to the loan application form.

Picture 2 – Loan Application Form



The image shows a web form titled "Smart Lender" with a house icon. The form is for "Loan Approval Prediction" and contains several input fields with dropdown menus. The fields are arranged in two columns. At the bottom, there is a large blue button labeled "Predict Loan Approval".

Smart Lender	
Loan Approval Prediction Form	
Gender Female	Married No
Dependents 0	Education Graduate
Self Employed Yes	Property Area Urban
Applicant Income 5000000	Coapplicant Income 2000000
Loan Amount 100000	Loan Amount Term 25000
Credit History Good (1)	
Predict Loan Approval	

Description:

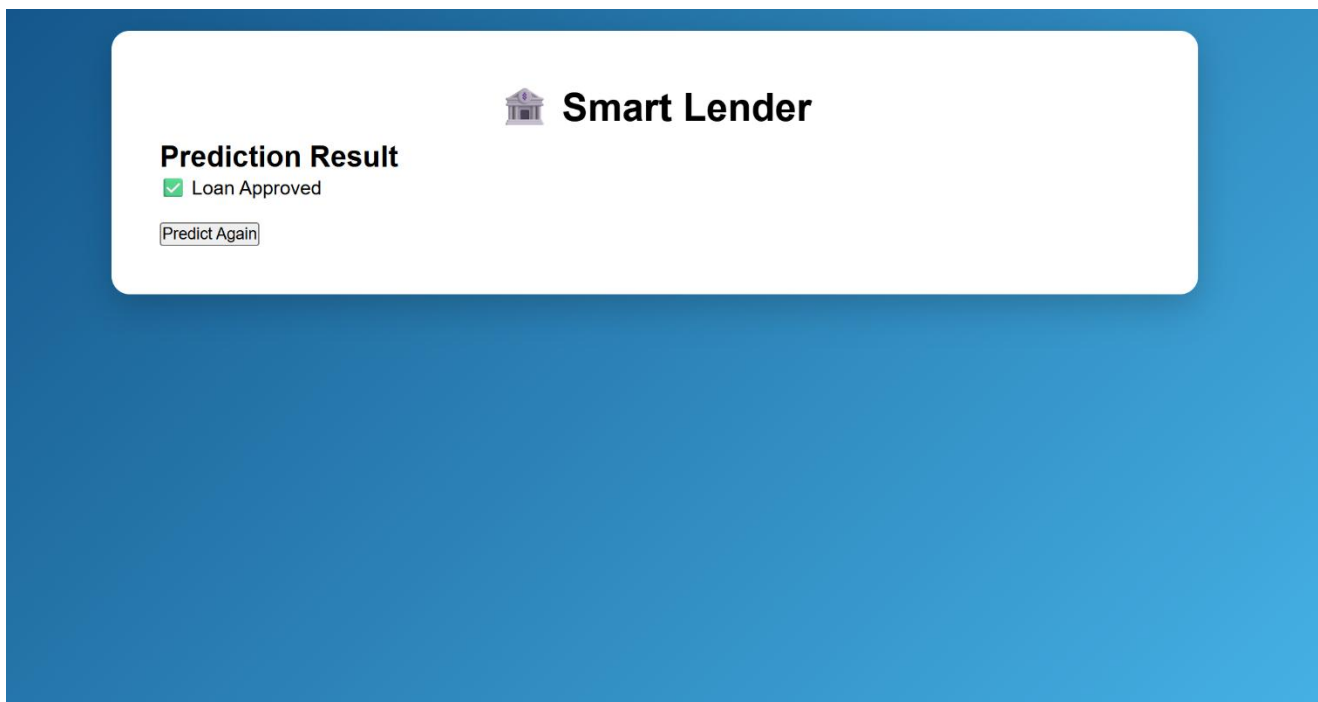
The application form allows users to enter all the required loan applicant information, including:

- Gender
- Marital Status
- Dependents
- Education
- Employment Status
- Property Area
- Applicant Income
- Co-applicant Income
- Loan Amount

- Loan Amount Term
- Credit History

After entering the required details, users can click the **Predict Loan Approval** button to submit the application.

Picture 3 – Prediction Result



Description:

After processing the submitted information, the trained machine learning model predicts the loan approval status. The application displays the prediction result (Loan Approved or Loan Rejected) and provides a **Predict Again** button, allowing users to submit another application without restarting the system.

Demo Scenario

1. Open the Smart Lender application.
 2. Click **Start Prediction**.
 3. Enter the applicant details in the loan application form.
 4. Click **Predict Loan Approval**.
 5. View the prediction result displayed by the machine learning model.
 6. Click **Predict Again** to test with different applicant information.
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Conclusion

The Smart Lender system successfully predicts loan approval using a trained machine learning model integrated with a Flask web application. The project demonstrates how artificial intelligence can assist financial institutions by providing fast, consistent, and reliable loan approval predictions through an easy-to-use web interface.